

3.5 Carbonyl Compounds

Question Paper

Course	CIEA Level Chemistry
Section	3. Organic Chemistry
Topic	3.5 Carbonyl Compounds
Difficulty	Hard

Time allowed: 20

Score: /10

Percentage: /100

Question 1

When HCN reacts with butanal, $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}$, product X is formed. If X is then hydrolysed by aqueous acid, product Y is produced.

When pentanal, $\text{C}_4\text{H}_9\text{CHO}$, is heated under reflux with acidified potassium dichromate (VI), product Z is formed.

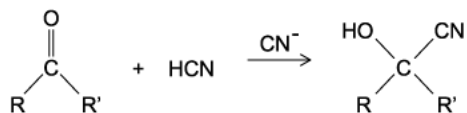
What is the difference in relative molecular mass of products Y and Z?

- A. 12
- B. 14
- C. 16
- D. 18

[1 mark]

Question 2

Cyanohydrins (also called hydroxynitriles) can be made from carbonyl compounds by generating CN^- ions from HCN in the presence of a weak base.



In a similar reaction, $^-\text{CH}_2\text{CO}_2\text{CH}_3$ ions are generated from $\text{CH}_3\text{CO}_2\text{CH}_3$ by strong bases.

Which compound can be made from an aldehyde and $\text{CH}_3\text{CO}_2\text{CH}_3$ in the presence of a strong base?

- A. $\text{CH}_3\text{CH}(\text{OH})\text{CO}_2\text{CH}_3$
- B. $\text{CH}_3\text{CO}_2\text{CH}_2\text{CH}(\text{OH})\text{CH}_3$
- C. $\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}_2\text{CO}_2\text{CH}_3$
- D. $(\text{CH}_3)_2\text{C}(\text{OH})\text{CH}_2\text{CO}_2\text{CH}_3$

[1 mark]

Question 3

When compound **T** reacts with its own oxidation product, a sweet-smelling liquid is produced.

What is the identity of compound **T**?

- A. Butanal
- B. Butanone
- C. Butan-1-ol
- D. Butanoic acid

[1 mark]

Question 4

Three separate test tubes, 1, 2 and 3, each contain 1 mol of compound **X**.

An excess of sodium borohydride is added to test tube 1 and an excess of acidified sodium permanganate is added to test tube 2. No reagent is added to test tube 3.

Following this, 2 mol of HCN (g) is added to each test tube and the HCN completely reacts in all 3.

What could **X** be?

- A. $\text{CH}_3\text{COCH}_2\text{CH}_2\text{CHO}$
- B. $\text{CH}_3\text{CH}(\text{OH})\text{COCH}_3$
- C. $\text{H}_2\text{C}=\text{CHCH}_2\text{CHO}$
- D. $\text{HOCH}_2\text{CH}_2\text{CH}_2\text{CHO}$

[1 mark]

Question 5

Glyceraldehyde, $\text{HOCH}_2\text{CH}(\text{OH})\text{CHO}$, is formed during photosynthesis on the pathway to glucose production.

Which reagents produce an organic product **without** a chiral carbon atom when reacted with glyceraldehyde?

- 1 Warmed acidified $\text{K}_2\text{Cr}_2\text{O}_7$
- 2 Fehling's reagent
- 3 NaBH_4

A. 1, 2 and 3

B. 1 and 3

C. 2 and 3

D. 1 only

[1 mark]

Question 6

Which of the following techniques will increase the rate of reaction between ethanal and aqueous hydrogen cyanide?

- 1 Increasing the temperature.
- 2 Performing the reacting under ultraviolet light.
- 3 Adding a small quantity of aqueous sodium cyanide.

A. 1 only

B. 1 and 3

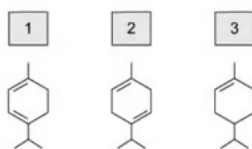
C. 2 and 3

D. 1, 2 and 3

[1 mark]

Question 7

Hydrocarbons 1, 2 and 3 all occur naturally.



All three are treated with hot acidified potassium manganate (VII) separately.

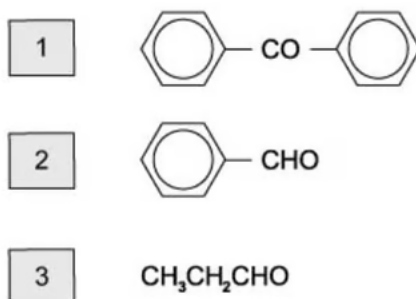
Which of the hydrocarbons 1, 2 and 3 will be split into two organic compounds, both containing a ketone group?

- A. 1 only
- B. 1 and 2
- C. 2 only
- D. 1 and 3

[1 mark]

Question 8

Which carbonyl compounds can be easily oxidised to carboxylic acids that are readily soluble in water?

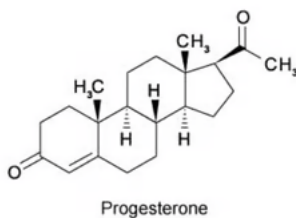


- A. 1 only
- B. 1 and 2
- C. 2 and 3
- D. 3 only

[1 mark]

Question 9

The hormone progesterone is produced during pregnancy to suppress ovulation.



Which reagents would give positive results with this compound?

- 1 Fehling's reagent
- 2 Aqueous bromine
- 3 2,4-dinitrophenylhydrazine

A. 1, 2 and 3

B. 1 and 2

C. 2 and 3

D. 1 only

[1 mark]

Question 10

The M_r of compound X is 72.

The composition by mass of X is 66.7 % carbon, 11.1 % hydrogen and 22.2 % oxygen.

X does not react with Fehling's reagent and gives an orange precipitate with 2,4-dinitrophenylhydrazine reagent.

What can be deduced from this information?

- 1 X is a ketone
- 2 X is butanone
- 3 X is a carbonyl compound

A. 1 and 2

B. 1, 2 and 3

C. 3 only

D. 1 only

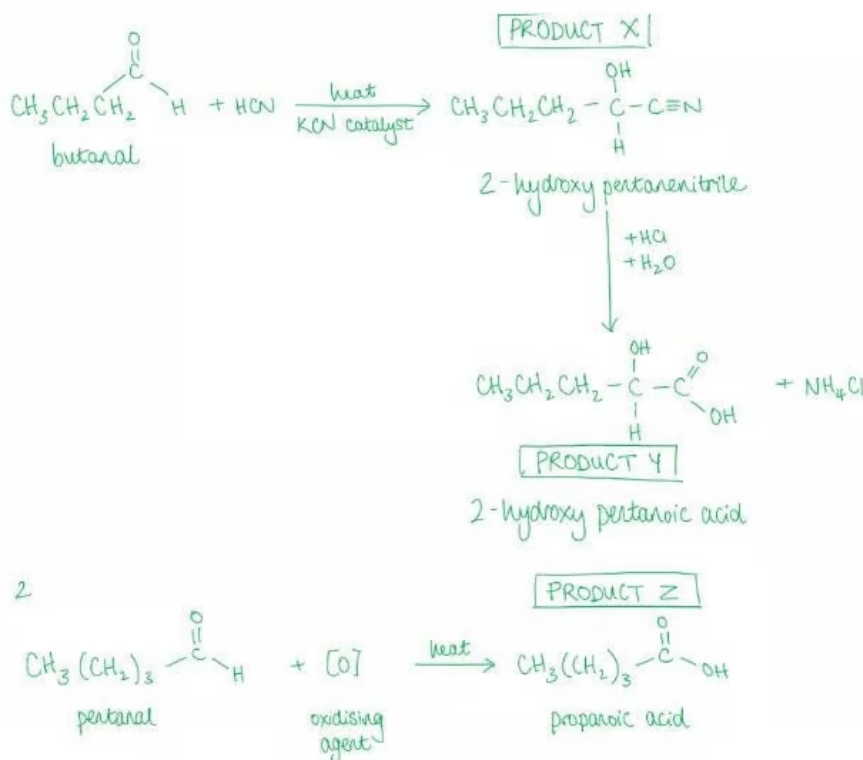
[1 mark]

Model Answers: Hard

1

The correct answer is **C** because:

- Butanal will undergo a nucleophilic addition reaction with HCN, KCN (catalyst) and heat to form the hydroxynitrile 2-hydroxypentanenitrile (product **X**).
- The **hydrolysis** of hydroxynitriles by aqueous acid, such as **dilute HCl**, will result in the formation of a **carboxylic acid** and an **ammonium salt**.
- **Hydrolysis** of 2-hydroxypentanenitrile gives 2-hydroxypentanoic acid (product **Y**).
- **Acidified potassium dichromate (VI)** is an **orange** oxidising agent which oxidises **primary alcohols, secondary alcohols** and **aldehydes**.
- **Pentanal** is an **aldehyde** and will get oxidised to pentanoic acid (product **Z**).



- The relative molecular masses are as follows:

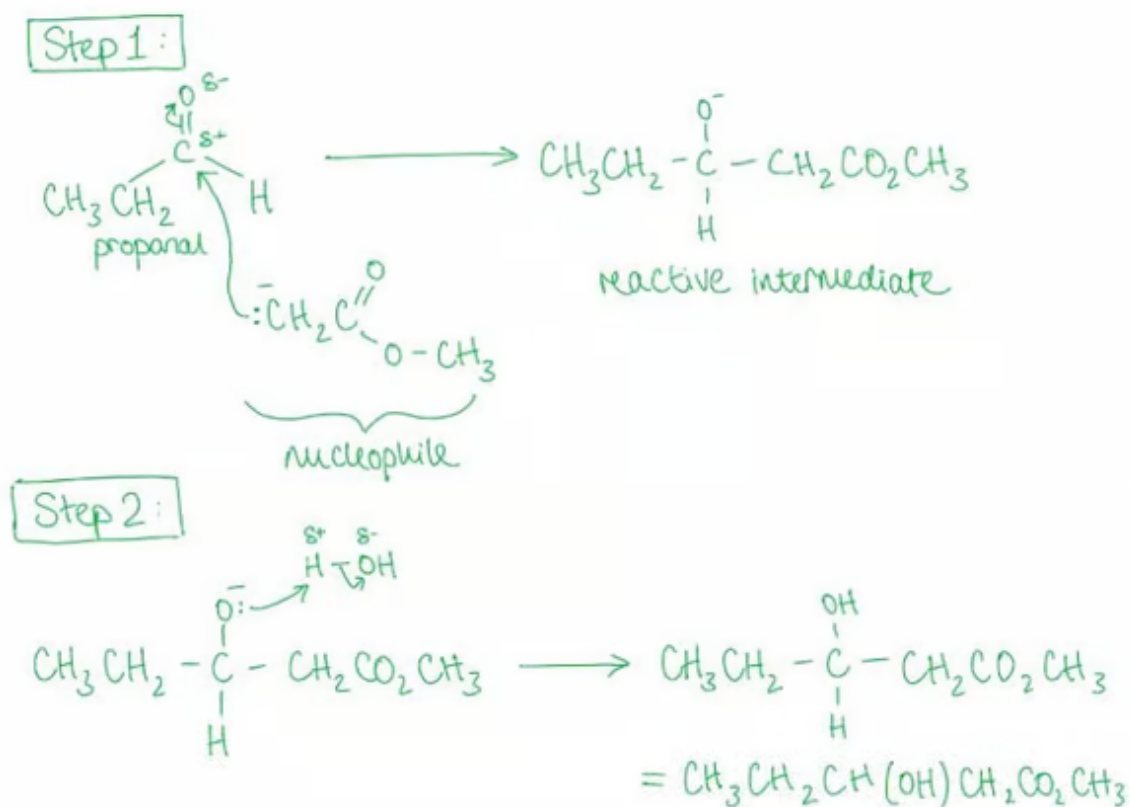
Compound	Carbon	Hydrogen	Oxygen	Total
Y ($\text{CH}_3(\text{CH}_2)_2\text{CH}(\text{OH})\text{COOH}$)	5 x 12	10 x 1	3 x 16	118
Z ($\text{CH}_3(\text{CH}_2)_3\text{COOH}$)	5 x 12	10 x 1	2 x 16	102

- The **difference** in relative molecular mass is $(118 - 102) = 16$

2

The correct answer is **C** because:

- Hydroxynitriles** can be formed from the **nucleophilic addition** reaction of **aldehydes** and **ketones** with **HCN**, **KCN** as catalyst and heat.
- $\text{CH}_3\text{CO}_2\text{CH}_3$ in the presence of a **strong base** will generate $^-\text{CH}_2\text{CO}_2\text{CH}_3$ which will act as a **nucleophile** in a similar way CN^- does.
- The reaction of propanal (**aldehyde**) with $\text{CH}_3\text{CO}_2\text{CH}_3$ in the presence of a strong base will form compound **C**.



A is incorrect as the aldehyde is **ethanal**. However, the **nucleophile** is not $^{-}\text{CO}_2\text{CH}_3$ but $^{-}\text{CH}_2\text{CO}_2\text{CH}_3$.

B is incorrect as the aldehyde is **methanal**. However, the **nucleophile** is not $^{-}\text{CHCH}_2\text{CO}_2\text{CH}_3$ but $^{-}\text{CH}_2\text{CO}_2\text{CH}_3$.

D is incorrect as this is a reaction between a **ketone** (propanone) and $^{-}\text{CH}_2\text{CO}_2\text{CH}_3$.

3

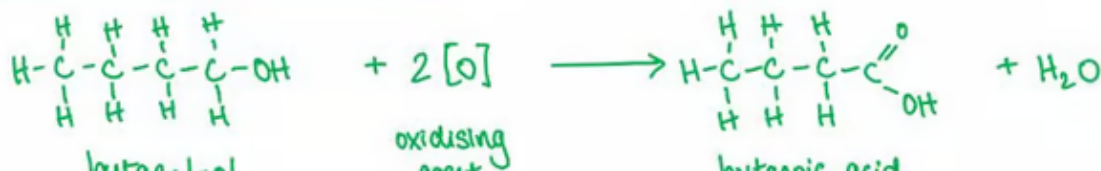
The correct answer is **C** because:

- **Esters** are **sweet** and **fruity smelling** organic compounds with a $-\text{COOR}$ group.
- **Esterification** is the reaction of **alcohols** with **carboxylic acids** (or **acyl chlorides**) to form esters.
 - The reagents are heated under reflux with a strong catalyst.
 - The reaction can be classified as a condensation reaction in which **two organic molecules** join together and **eliminate** a small molecule.
- Butan-1-ol is a **primary alcohol** which can be oxidised to butanoic acid (a

carboxylic acid).

- The reaction of butan-1-ol with butanoic acid will form the ester **butyl butanoate**.

Oxidation of Compound T





A is incorrect as butanal is an **aldehyde** and will get oxidised to **butanoic acid**. The reaction of butanal with butanoic acid will **not** produce a sweet-smelling liquid ester.

B is incorrect as butanone is a **ketone** and cannot get further oxidised.

D is incorrect as butanoic acid is a **carboxylic acid** and cannot get further oxidised.

4

The correct answer is **B** because:

- **Hydrogen cyanide** will react with **alcohols** and **carbonyl** compounds.
- Therefore, for the same number of moles of HCN to react in all three test tubes the total number of alcohols and carbonyl groups combined must be unchanged.
- **1 mol HCN** reacts with **1 mole of alcohol or carbonyl**, therefore if 2 mol HCN is reacting with a single molecule there must be 2 alcohol / carbonyl groups present in that molecule.
- In option B there is a secondary alcohol group $\text{CH}_3\text{CH}(\text{OH})\text{COCH}_3$ and a ketone $\text{CH}_3\text{CH}(\text{OH})\text{COCH}_3$ both of which will each react with 1 HCN = **2 total**.
- In test tube 1 the **reducing agent** NaBH_4 will cause the **ketone** to reduce to a secondary alcohol resulting in 2 secondary alcohol groups = **2 total**.
- In test tube 2 the **oxidising agent** MnO_4^- will oxidise the **secondary alcohol** to a ketone but will not oxidise the ketone further, resulting in 2 ketone groups = **2 total**.

A is incorrect as in test tube **2** the **aldehyde** in this molecule will **oxidise** to a carboxylic acid which will not react with HCN, leaving only the ketone = **1 total**.

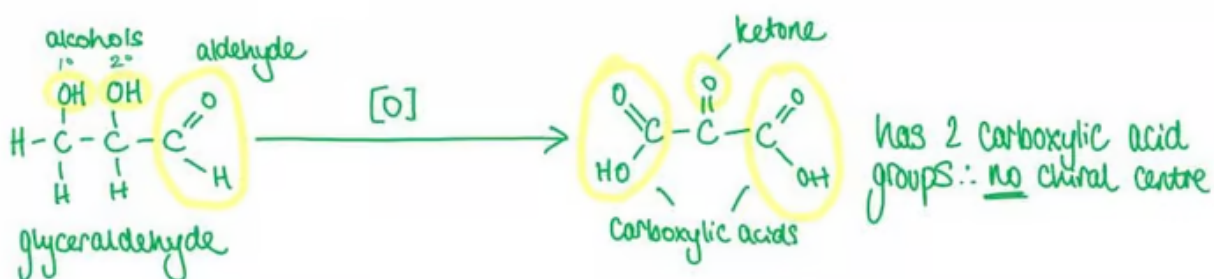
C is incorrect as this molecule contains an **alkene**, which does not react with HCN, meaning this molecule only has the aldehyde that will react = **1 total**. In test tube **2** the **aldehyde** is **oxidised** to a carboxylic acid = **0 total**.

D is incorrect as in test tube **2** the **aldehyde** and **primary alcohol** in this molecule will **oxidise** to carboxylic acids which don't react with HCN = **0 total**.

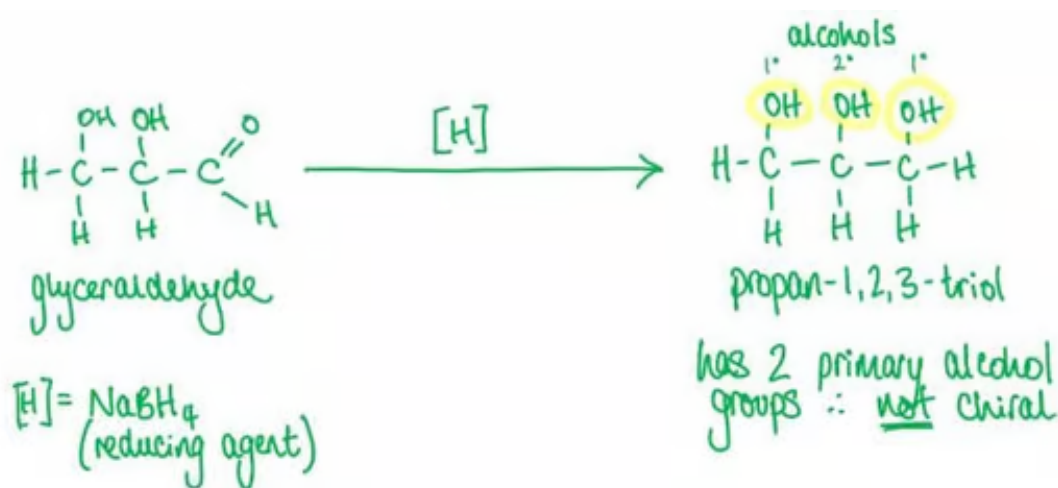
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The correct answer is **B** because:

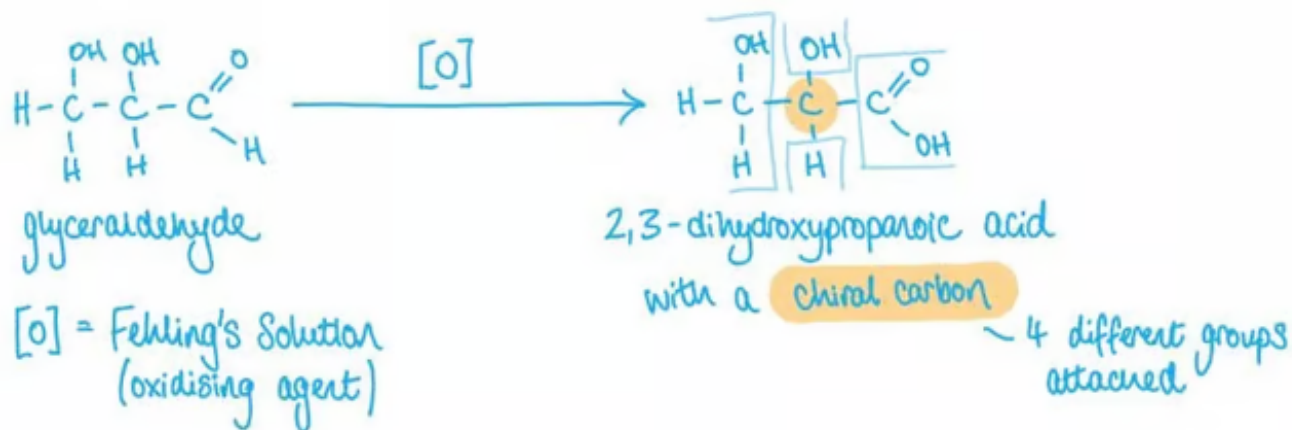
- **Warm acidified potassium dichromate (VI)** is an oxidising agent which oxidises primary alcohols, secondary alcohols and aldehydes.
- Glyceraldehyde contains a **primary alcohol**, a **secondary alcohol** and an **aldehyde** which can get oxidised to a carboxylic acid, a ketone and a carboxylic acid respectively.
- The product of this oxidation reaction does **not** contain any chiral centres.



- **NaBH_4** is a reducing agent which will reduce the **aldehyde** to a **primary alcohol**.
- The product of this reduction reaction does **not** contain any chiral centres.



A & C are incorrect as Fehling's solution **oxidises** the **aldehyde** to a **carboxylic acid** to give a **chiral** product, Fehling's does not oxidise alcohol groups.



6

The correct answer is **B** because:

- Increasing the temperature causes a greater proportion of particles to have the activation energy.
- The particles move faster and there is an increase in **effective collision frequency**.
- Aqueous sodium cyanide is a **catalyst**.
- A catalyst increases the rate of forward and backward reaction equally without taking part in the reaction itself.

A is incorrect as adding a small amount of sodium cyanide will also increase the rate of reaction.

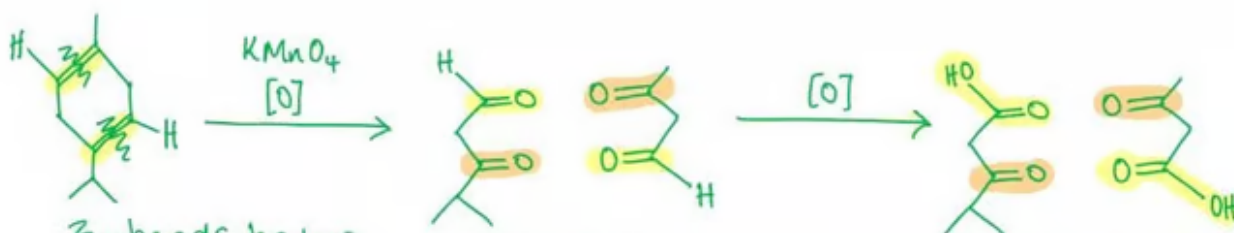
C & D are incorrect as this is not a **free-radical substitution** reaction so the use of UV light will not increase the rate of reaction.

7

The correct answer is **C** because:

- Oxidation** in organic chemistry is the gain of oxygen atoms.
- The acidified potassium manganate (VII) solution **oxidises** the alkene by **breaking** the C=C double bond and replacing it with two C-O double bonds.
- The products are known as **carbonyl compounds**.

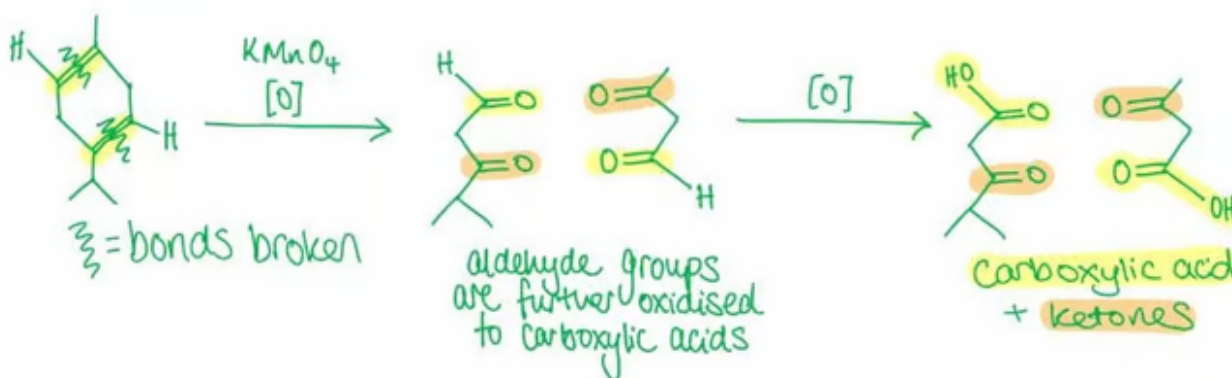
- In compound **2**, **strong** oxidation with KMnO_4 results in the formation of two products each containing **one ketone** and **one aldehyde group**.
 - The aldehydes will get further oxidised to carboxylic acids.



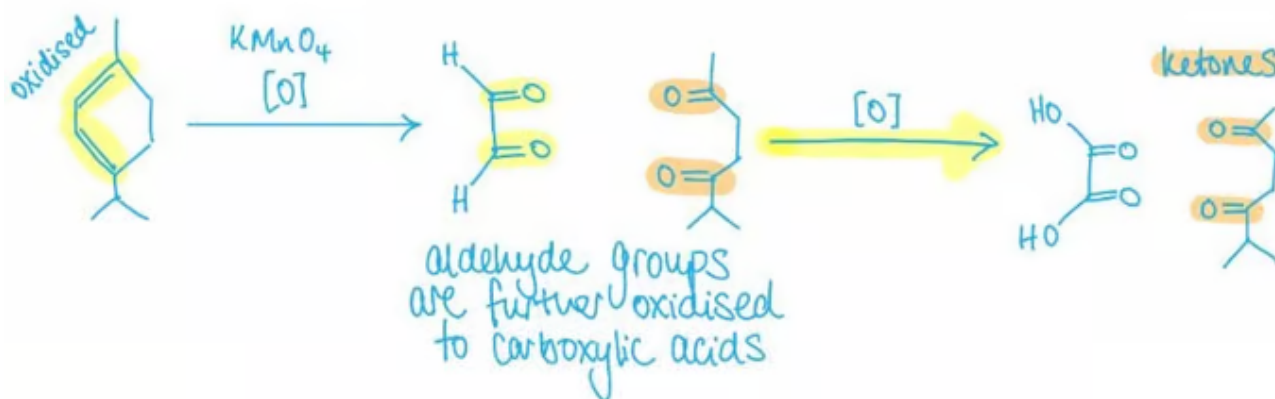
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The correct answer is **C** because:

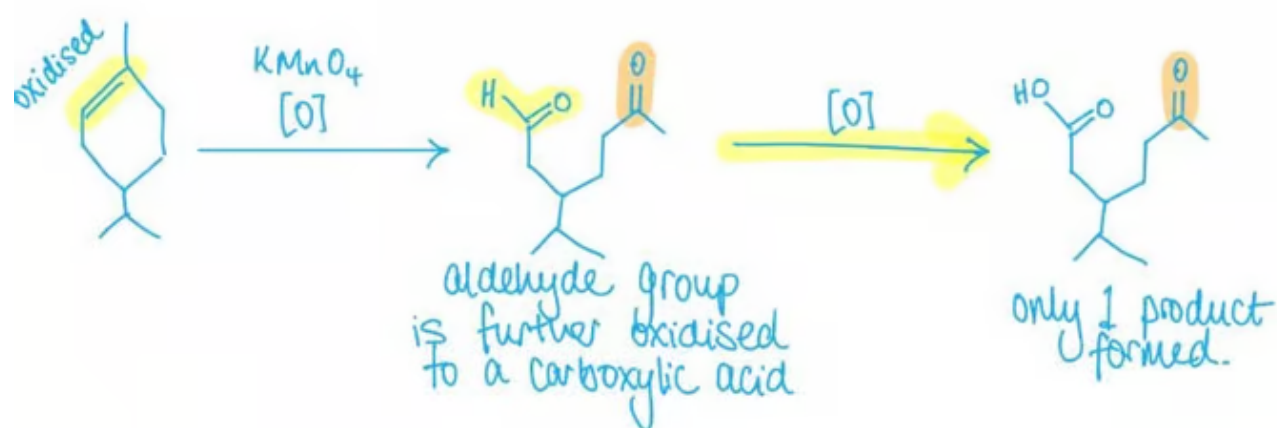
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- In compound **2**, **strong** oxidation with KMnO_4 results in the formation of two products each containing **one ketone** and **one aldehyde group**.
 - The aldehydes will get further oxidised to carboxylic acids.



A & B are incorrect as two oxidation products are formed; one with **no** ketone groups and one with **two** ketone groups.



D is incorrect as one product is formed which contains one ketone group only.



8

The correct answer is **D** because:

- Compound **3** is an **aldehyde** (ethanal) which can be oxidised to a carboxylic acid (ethanoic acid).
- **Ethanoic acid** contains a polar carboxylic acid group ($-\text{COOH}$) which can form hydrogen bonds with water.
- Therefore, compound **3** can be oxidised to a carboxylic acid which is readily soluble in water.

A & B are incorrect as compound **1** contains a **ketone group** which **cannot** be oxidised.

C is incorrect as compound **2** contains an **aldehyde** group which can be oxidised to a carboxylic acid. However due to the **non-polar benzene ring**, the product (**benzoic acid**) is **not** readily soluble in water.

9

The correct answer is **C** because:

- **Alkenes** are saturated molecules that will react with aqueous bromine.
- **Bromine water** (which is $\text{Br}(\text{aq})$) is an **orange** or **yellow** solution.
- When progesterone is shaken with the bromine water, an **electrophilic addition** reaction will take place and the solution will decolourise over time.

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- The Br_2 will add on to the double bond in the alkene.
- The presence of a **carbonyl group** in an **aldehyde** or **ketone** can be tested by adding a solution of 2,4–dinitrophenylhydrazine (**2,4–DNPH**).
- If an aldehyde or ketone is present, a **deep–orange precipitate** is formed.
- As progesterone contains alkene and ketone functional groups, aqueous bromine and 2,4–dinitrophenylhydrazine will give positive results.

A, B & D are incorrect as the Fehling's solution is an **alkaline** solution containing copper (II) ions. When warmed with an **aldehyde**, the Cu^{2+} ions are reduced to Cu^+ ions changing the clear **blue** Fehling's solution colour to an opaque red / orange colour; as the precipitate of copper (I) oxide forms. Progesterone contains **no** aldehyde groups and will therefore give no positive result with Fehling's solution.

10

The correct answer is **B** because:

- The presence of a **carbonyl group** in an **aldehyde** or **ketone** can be tested by adding a solution of 2,4–dinitrophenylhydrazine (**2,4–DNPH**).
- If an aldehyde or ketone is present, a **deep–orange precipitate** is formed.
- Therefore, **X** can be either an **aldehyde** or a **ketone**.
- Since, compound **X** does not react with Fehling's reagent, compound **X** has to be a **ketone**.
- A ketone is a **carbonyl compound**.
- The composition of compound **X** is:

Atom	Mass	Number of atoms
Carbon	$72 \times 0.667 = 48$	$\frac{48}{12} = 4$
Hydrogen	$72 \times 0.111 = 8$	$\frac{8}{1} = 8$
Oxygen	$72 \times 0.222 = 16$	$\frac{16}{16} = 1$

oxygen	12 x 0.222 = 10	16
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- So, compound **X** is C_4H_8O corresponds to the ketone **butanone**.
- Therefore, statements **1** and **3** can be deduced from the above information.

